

Module Lattice Based Key Encapsulation Mechanism Standard (ML-KEM)

ML-KEM is a lattice based key encapsulation mechanism standardized by NIST's FIPS 203 publication that provides a quantum resistant method of establishing shared secret keys between parties communicating over a public channel. It's set to replace current key encapsulation algorithms such as RSA (Rivest-Shamir-Adleman) and Diffie-Hellman.

ML-KEM based on the hardness of the MLWE (Module Learning With Errors) problem, which is itself based on the idea of inferring a linear function over noisy data. The problem is considered to be quantum resistant, which carries over to ML-KEM.